

Content-adaptive sparse attention: a synthetic research illustration

Abstract

A fixed sparse attention pattern spends the same budget on every input, although the useful connections may vary with content. This note formulates content-adaptive block selection and illustrates a matched-budget comparison using nine synthetic measurements. At a retained fraction of 0.25, mean illustrative accuracy is 88% for content-adaptive selection and 86% for a fixed pattern; the dense reference is 90%. These authored values demonstrate the structure of a research comparison and do not establish performance on a real model or dataset.

Research question

The question is whether content-dependent allocation preserves more task information than a fixed allocation at the same sparsity. Separating allocation from budget is essential: a comparison with different retained fractions cannot isolate the selection mechanism.

Candidate method

Partition the attention matrix into blocks. For block b , compute a relevance score s_b from its query and key content. Given B blocks and a retained fraction ρ , retain $K = \text{ceil}(\rho B)$ blocks with the highest scores. Normalize attention within the retained entries. The fixed sparse control retains the same K blocks according to a content-independent pattern.

Demonstration comparison

The CSV pairs three illustrative seeds across dense, fixed sparse and content-adaptive conditions. Accuracy is measured in percent. Each condition has a sample standard deviation of one percentage point.

Method	Retained fraction	Mean accuracy (%)
Dense	1.00	90
Fixed sparse	0.25	86
Content-adaptive	0.25	88

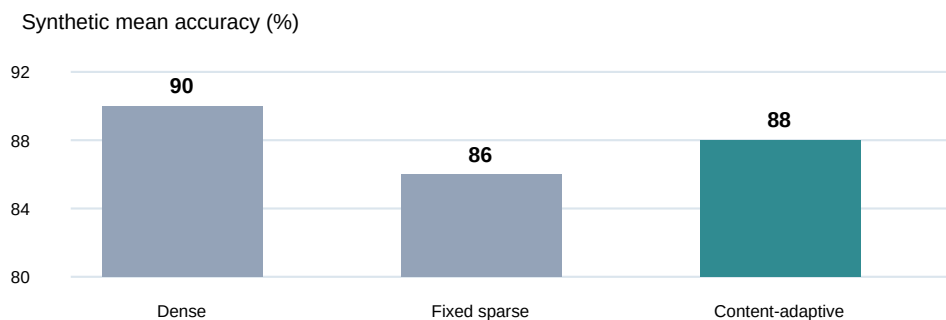
The paired difference between the two sparse conditions is two percentage points in every row. Content-adaptive accuracy remains two percentage points below the dense reference. Figure 1 presents these same means; `code/analyze.py` calculates them directly from `data/pilot.csv`.

Evidence boundary

All measurements are synthetic demonstration data. They support arithmetic inspection of the example, not an empirical advantage. A real study would include a random-selection control, multiple retention budgets, end-to-end latency, peak memory and genuine repeated training runs. The selector's overhead cannot be inferred from the retained fraction.

Conclusion

The comparison separates the selection rule from its budget and gives each proposed claim a corresponding test. Its numerical pattern is illustrative; the content-allocation hypothesis remains untested on real workloads.



内容自适应稀疏注意力：合成研究示例

摘要

固定稀疏注意力对每个输入采用相同的预算分配，但有效连接可能随内容变化。本札记提出按内容选择注意力块的候选机制，并以九条合成数据说明同预算比较。在保留比例为 0.25 时，内容自适应选择的演示平均准确率为 88%，固定模式为 86%，稠密参照为 90%。这些编写的数值展示研究比较的组织方式，不证明真实模型或数据集上的性能。

研究问题

问题是：相同稀疏度下，按内容分配预算能否比固定分配保留更多任务信息？必须区分分配规则与预算大小；若保留比例不同，就无法单独识别选择机制的作用。

候选方法

将注意力矩阵划分为块。对块 b ，根据查询与键的内容计算相关性得分 s_b 。给定 B 个块和保留比例 ρ ，保留得分最高的 $K = \text{ceil}(\rho B)$ 个块，并在保留条目内归一化注意力。固定稀疏对照使用与内容无关的模式保留相同数量 K 的块。

演示比较

CSV 将三个演示种子在稠密、固定稀疏、内容自适应条件间配对。准确率单位为百分比，每个条件的样本标准差均为 1 个百分点。

方法	保留比例	平均准确率 (%)
稠密	1.00	90
固定稀疏	0.25	86
内容自适应	0.25	88

两个稀疏条件在每个配对行中相差 2 个百分点。内容自适应仍比稠密参照低 2 个百分点。图 1 展示同一组均值；code/analyze.py 从 data/pilot.csv 直接计算这些结果。

证据边界

全部数值都是合成演示数据，仅用于检查示例中的计算，不构成实证优势。真实研究需补充随机选择对照、多种预算、端到端延迟、峰值显存和真正重复的训练。不能从保留比例推断选择器的运行开销。

结论

此比较将选择规则与预算分开，并为每项拟议结论安排相应检验。数值模式仅供演示；内容分配假设尚未在真实任务中得到验证。

